Turpu Saandeep Sai

Backend & AI Systems Developer

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Summary

Backend-focused software engineer who designs and ships async, multi-stage AI pipelines end to end – from API layer to background job orchestration to third-party AI service integration. As a freelance developer, delivered full-stack applications for external clients alongside personal SaaS projects, including a Django backend that generates full educational videos using rate-limited LLM calls, custom caching, and cloud deployment.

Education

B.Tech in Artificial Intelligence & Data Science, CGPA: 8.3 *2023 – 2027 (Expected)*
Sri Indu College of Engineering and Technology

Associate of Science in Artificial Intelligence *2025 – Present*
Saras AI Institute

Space Junior College *2021 – 2023*
XII (State Board)Score: 93.9%

SR Digi School *2020 – 2021*
X (SSC)Score: 95%

Projects

Educational Video Generator – SaaS (Personal) *GitHub · Demo*

- Architected a Django backend that turns a topic into a fully rendered educational video: LLM-generated script → Manim animation → TTS voiceover → FFmpeg audio/video sync → cloud upload.
- Engineered an async job system with background workers and a polling status endpoint to handle 10–13 minute generation jobs without blocking the API layer.
- Implemented a token-bucket rate limiter with API key rotation, a 7-day LRU cache, and an 8-worker parallel rendering pipeline; deployed to a cloud host with a dedicated health-check endpoint.

SafeSteps (TrustHut) – Community Road Safety Platform (Client Project) *Live Demo · GitHub*

- Developed the frontend and integrated with a Django REST backend for a platform where users report accessibility and road-hazard locations on an interactive map.
- Designed a route optimizer that compares multiple paths and scores them by community-reported hazard density.
- Integrated an LLM-powered safety chatbot alongside Firebase authentication and a highway-risk heatmap.

College Placement Guide (Client Project) *Live Demo · GitHub*

- Built a placement-management system connecting students, faculty/TPO, and placement drives on a Node/Express + Firebase Admin SDK backend.
- Integrated an LLM-based resume analyzer (Llama 3.1 via OpenRouter) that scores resumes for ATS compatibility and generates company-specific resume variants.
- Built the faculty dashboard: drive management, bulk student data import, and placement analytics.

Audiobook Generator (Personal) *GitHub*

- Automated full audiobook generation from a topic: LLM-structured narrative segments with emotion-based TTS voice selection and cinematic background mixing.
- Designed a chunked, parallelized generation pipeline to avoid API timeouts on long-form narration.

Core Technologies

- **Languages:** Python, JavaScript, Java, SQL
- **Backend:** Django, Django REST Framework, FastAPI, Node.js, Express, REST API design, async/background job processing
- **AI Engineering:** LLM API integration (Gemini, Llama 3.1 via OpenRouter), prompt-driven content generation, TTS pipelines, rate limiting & API key rotation, caching strategies
- **Frontend:** React, Next.js, Vite, TypeScript
- **Cloud, DevOps & Data:** Docker, AWS, Git/GitHub, cloud deployment (Render.com), Firebase (Auth, Firestore, Admin SDK)

- **Media Processing:** Manim, FFmpeg, Edge TTS

Non-Technical Skills

- Confident technical communication
- Team leadership & coordination
- Working effectively under deadline pressure

Achievements

- **1st Prize** – Pitch Arena 2026 (Sri Indu College of Engineering and Technology)
- **1st Prize** – Invenza Ideathon 2026 (Geethanjali College of Engineering and Technology)
- **2nd Prize** – Nexovate 2025 (Geethanjali College of Engineering and Technology)
- **2nd Prize** – Cypher 2025 (CVR College of Engineering)
- **2nd Prize** – Jijnasa 2025 & 2026 (Bhoj Reddy Womens College)